Joint calibration of roughness, leverage, Taylor, Zumbach and excess kurtosis for the fBm-bank ESN with bounded readouts and a modulated rough bank

At the optimal architecture of the grid-search protocol; per-asset calibration on 3 indices and 6 stocks

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Abstract

We apply the joint-calibration protocol of the companion note (v4) to the continuous-time fBm-bank ESN with bounded non-parametric readouts and a Lévy-modulated rough bank:

$$\sigma_t = \tilde{\sigma}(t)\, e^{\psi(\zeta_t)} F(P_t), \qquad \tilde{\sigma}(t) = ae^{-bt} + c(1 - e^{-bt}).$$

The architecture is fixed at the optimum of the grid-search protocol and identical for every asset:

- rough bank $M = 18$, $H = 0.0577$;
- modulation $\lambda_\Gamma = 2.47$, $\Lambda = 4.63$, $a_J = 1.06$;
- leverage window $\Theta \approx 37$ days;
- feedback bank $J = 2$;
- readout pieces $(N_c, N_d, N_e) = (5, 4, 1)$.

Per asset, the 17 parameters of σ, dX and the initial state are calibrated jointly on the note's 7-block, 64-term objective:

$$\theta = (c_1..c_5,\ d_1..d_4,\ e_1,\ a,\ b,\ c,\ \varrho,\ \zeta_0,\ P_0,\ \Gamma_0),$$

covering roughness, leverage at lags 1–40, the two Taylor curves, the two Zumbach curves and excess kurtosis. The bench has three indices (S&P 500, Russell 2000, Nasdaq-100) and six stocks (GPC, QCOM, INTC, LUV, NI, KMB).

With $K = 64$ calibration paths, the validation objective on 20 independent replicates is 0.24–0.59 for eight assets and 1.32 for the S&P 500. This is lower than in the previous version of this calibration (two-slope readout, $(1 + P^2)^\gamma$, unmodulated bank) for seven of the nine assets. A first run with $K = 16$ overfitted the common noise, which motivated $K = 64$. The real target lies in the model's 90% band for:

- $\hat{H}$ and kurtosis: 9/9 assets;
- leverage: 94% of lags (96% over lags -40 to 40);
- ACF$|x|$: 87%; ACFx^2: 67%; their difference: 93%;
- the Zumbach curves: 91% and 91%; their difference: 98%.

The Zumbach asymmetry of the stocks, too weak in the previous version, roughly doubles but remains below the market's. The main misfits are:

- a Taylor difference that is too large for the S&P 500 and Nasdaq-100;
- same-day leverage that is too strong;
- a roughness that is slightly too low for the S&P 500.

Many parameters end on a bound, notably the weakly identified level (a, c) and initial state.

The calibration was stopped after 7 (first five assets) and 9 (last four) of its 25 iterations, with the objective still decreasing slowly.

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1 Set-up

1.1 Assets and data

- **Indices**: GSPC (S&P 500), RUT (Russell 2000), NDX (Nasdaq-100), with daily OHLC from the index file `indices_data_all_quotations1990-2023.xlsx`.
- **Stocks**: GPC, QCOM, INTC, LUV, NI (the assets of the grid-search protocol) and KMB, from the S&P 500 constituent file `list_dics_SP500_ohlcvol_19940101_filtered.pkl`.

For each asset, the last $8\,052 = 6 \times 1\,342$ trading days are split into 6 windows of $T = 1\,342$ days, the length of every simulated path, and $x_t = \log(C_t/C_{t-1})$. The index file contains exactly three series with daily open, high, low and close prices, so the ninth asset is a sixth stock, KMB (Kimberly–Clark). All nine assets have real OHLC data and are measured with the same Garman–Klass proxy.

1.2 Market windows

A **market window** is one consecutive block of $T = 1\,342$ trading days (about 5.3 years) of an asset's real history.

Table 1: The six market windows. The index file ends in November 2023; the constituent file (stocks) runs to December 2025.

window	stocks	GSPC, RUT, NDX
1	1994-01-05 → 1999-04-29	1991-11-26 → 1997-03-18
2	1999-04-30 → 2004-08-31	1997-03-19 → 2002-07-19
3	2004-09-01 → 2009-12-29	2002-07-22 → 2007-11-15
4	2009-12-30 → 2015-04-30	2007-11-16 → 2013-03-19
5	2015-05-01 → 2020-08-27	2013-03-20 → 2018-07-17
6	2020-08-28 → 2025-12-31	2018-07-18 → 2023-11-14

1. Each window yields one value of every statistic of Section 3, computed exactly as on one simulated path.
2. The **market target** of a statistic is the mean of its six window values; this is what the calibration fits.
3. The six window values are also kept. Their spread is the market's own variability over time; the model's 90% band should be read against it.

Windows of the path length make the comparison like for like. $\hat{H}$ depends on T through its lag range $[1, T/4]$, and raw kurtosis grows with the sample length for heavy-tailed returns.

How to read the figures. In every model-vs-market figure:

- **light red lines** (light red dots for $\hat{H}$) are the six individual windows;
- **bold red dots** (red diamond for $\hat{H}$) are their mean, i.e. the target;
- the **green line** is the mean over 20 simulated validation paths of 1 342 days;
- the **green band** is the 5%–95% range across those paths.

2 The model in continuous time

Time is in years ($\delta = 1/252$ per trading day). Every path starts at $t = 0$. On $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$:

- $W^1, \dots, W^M, W^\perp$ are independent Brownian motions, and $W = (W^1, \dots, W^M)^\top$;
- L is an independent compound Poisson process with intensity Λ and $\mathrm{Exp}(a_J)$ jump sizes.

Layer 0′: vol-of-vol modulation.

$$d\Gamma_t = -\lambda_\Gamma \Gamma_t \, dt + dL_t, \qquad \Gamma_0 > 0. \tag{2.1}$$

Layer 1: modulated rough bank. With $\lambda_i = \lambda_{lo}(\lambda_{hi}/\lambda_{lo})^{(i-1)/(M-1)}$,

$$dr_t^i = -\lambda_i r_t^i \, dt + \sqrt{2\lambda_i \Gamma_t} \, dW_t^i, \qquad \zeta_t = \frac{q^\top r_t}{\sqrt{\gamma_0}}, \tag{2.2}$$

$$q_i = \sqrt{b_i}, \qquad b = \arg\min_{b \geq 0} \int_{\tau_{\min}}^{\tau_{\max}} \Big(\frac{2 \sum_i b_i (1 - e^{-\lambda_i \tau})}{\tau^{2H}} - 1 \Big)^2 \frac{d\tau}{\tau}, \qquad \gamma_0 = \sum_i b_i.$$

Layer 0: return noise and leverage.

$$dW_t^X = \varrho \, u^\top dW_t + \sqrt{1 - \varrho^2} \, dW_t^\perp, \qquad u = \frac{m}{\|m\|}, \quad m_i = q_i \sqrt{2\lambda_i} \, \frac{1 - e^{-\lambda_i \Theta}}{\lambda_i \Theta}.$$

Layer 2: endogenous feedback bank. For $j = 1, \dots, J$,

$$ds_t^j = -\kappa_j s_t^j\, dt + \sqrt{2\kappa_j}\, \frac{\sigma_t}{\tilde{\sigma}(t)}\, dW_t^X, \qquad P_t = w^\top s_t, \qquad \kappa_j = \kappa_{hi}\Big(\frac{\kappa_{lo}}{\kappa_{hi}}\Big)^{\frac{j-1}{J-1}}, \quad \|w\| = 1.$$

Readout, volatility and price.

$$\sigma_t = \tilde{\sigma}(t)\, e^{\psi(\zeta_t)}\, F(P_t), \qquad \tilde{\sigma}(t) = a\, e^{-bt} + c\,(1 - e^{-bt}), \qquad dX_t = -\tfrac{1}{2}\sigma_t^2\, dt + \sigma_t\, dW_t^X. \tag{2.3}$$

- ψ is continuous and piecewise linear on the $N_c + 1$ equispaced knots of $[-3, 3]$, with slope $-c_m$ on piece m, $\psi(0) = 0$, and constant beyond ± 3.
- F is continuous and piecewise linear on the knots $p_{-N_d} < \cdots < 0 < \cdots < p_{N_e}$, with $F(0) = 1$, increments $d_j \geq 0$ on the loss side and log-decays $e_j \geq 0$ on the gain side, and constant beyond the outer knots. The knot rule is that of the grid-search documentation, Definition 2.1.

Initial state. Given (ζ_0, P_0, Γ_0) and $g \sim \mathcal{N}(0, I_M)$,

$$r_0 = \sqrt{\Gamma_0}\, g - q\, \frac{q^\top \sqrt{\Gamma_0}\, g - \zeta_0 \sqrt{\gamma_0}}{\|q\|^2}, \qquad s_0 = P_0\, w, \tag{2.4}$$

the stationary law at level Γ_0, conditioned on $q^\top r_0 = \zeta_0 \sqrt{\gamma_0}$.

Measurement layer. Day n consists of an overnight interval of length $\phi\delta$ and a session $S_n = [t_n + \phi\delta, t_{n+1}]$. The observations are $x_n = X_{t_{n+1}} - X_{t_n}$ and

$$v_n = \tfrac{1}{2}\big(\sup_{S_n} X - \inf_{S_n} X\big)^2 - (2\ln 2 - 1)\big(X_{t_{n+1}} - X_{t_n + \phi\delta}\big)^2 \qquad \text{(Garman–Klass, all assets).}$$

2.1 Fixed architecture (all assets)

Table 2: Architecture, fixed at the optimum of the grid-search protocol and identical for every asset.

M	18	J	2
H	0.0577	κ_{lo}, κ_{hi} (yr^{-1})	1.008, 80.22
λ_{lo}, λ_{hi} (yr^{-1})	0.001, 231.9	time scales $1/\kappa_j$	3.1, 250 days
Θ (days)	36.8	w	(0.995, -0.099)
λ_Γ, Λ (yr^{-1})	2.467, 4.628	a_J (mean jump $1/a_J$)	1.063 (0.941)
(N_c, N_d, N_e)	(5, 4, 1)	γ_0, variogram fit error	0.679, 24%

Per asset, fixed before calibration. The overnight share ϕ (Table 6).

3 Methodology

3.1 The calibrated parameter vector

Per asset, all 17 parameters of σ, dX and the initial state are calibrated, inside the box of the grid-search protocol, i.e. between the two levels of its parameter design:

$$\theta = \big(c_1, \dots, c_5,\ d_1, \dots, d_4,\ e_1,\ a,\ b,\ c,\ \varrho,\ \zeta_0,\ P_0,\ \Gamma_0\big) \in \mathbb{R}^{17}.$$

Table 3: Calibrated parameters and their bounds. Every asset starts at the centre of the box.

parameter	low	high	role
c_1, c_2	0.3	0.7	slopes of $-\psi$ on $[-3, -1.8]$, $[-1.8, -0.6]$
c_3	0.25	0.65	slope of $-\psi$ on $[-0.6, 0.6]$
c_4, c_5	0.2	0.6	slopes of $-\psi$ on $[0.6, 1.8]$, $[1.8, 3]$
$d_1, \dots, d_4$	0	0.3	loss-side increments of F on the knots $-8, -5, -3, -1.5, 0$
e_1	0	0.9	gain-side log-decay of F on $[0, 5]$
a	0.15	0.25	level of $\tilde{\sigma}$ at $t = 0$
b (yr^{-1})	0.5	5	speed of $\tilde{\sigma}$
c	0.15	0.25	long-run level of $\tilde{\sigma}$
ϱ	0.2	0.7	leverage correlation
ζ_0, P_0	-1.5	1.5	initial rough and feedback states
Γ_0	0.3	1.5	initial modulation

Proposition 3.1 (Weakly identified parameters)**.** *1.* ***Level.*** *Scaling $(a, c) \mapsto (ka, kc)$ scales $\tilde{\sigma}(t)$, hence σ, by k. It leaves ζ, P and Γ unchanged, since Layer 2 is driven by $(\sigma/\tilde{\sigma})\, dW^X$. Neglecting the convexity drift $-\frac{1}{2}\sigma^2$, $x' = kx$ and $v' = k^2 v$; hence $\hat{H}$, every correlation and the excess kurtosis are unchanged. Only the ratio a/c (and b) enters the targets.*

2. ***Initial state.*** *(ζ_0, P_0, Γ_0) act on a path through a transient. It decays at the rates λ_i (rough bank), κ_j (feedback) and λ_Γ (modulation). With the architecture's rates, ζ_0 and P_0 are forgotten within weeks, and Γ_0 within $1/\lambda_\Gamma \approx 0.4$ years, i.e. within 8% of a 1 342-day window.*

Proof. (1) ζ, Γ and P do not depend on the level: Layers $0'$ and 1 do not involve σ, and the driver of Layer 2 is $\sigma/\tilde{\sigma} = e^{\psi} F$. Then $\sigma' = k\sigma$ and $dX' = k\, dX$ up to the drift, and the Garman–Klass proxy is 2-homogeneous in the path. $\hat{H}$ uses differences of $\log v$; correlations and kurtosis are scale-free. (2) The initial values enter the solutions only through the factors $e^{-\lambda_i t}$, $e^{-\kappa_j t}$ and $e^{-\lambda_\Gamma t}$. □

Note. The objective carries little information on the level (a, c) and on the initial state. As in the previous version of this calibration, these parameters are calibrated as specified, but their final values are largely set by the residual noise of the objective, and several end on a bound. The model's mean volatility is therefore not matched to the market's. A level target would identify it.

3.2 Statistics

Per path (or per real window):

- $\hat{H}$: $\frac{1}{2}$ times the OLS slope of $\log d(L)$ on $\log L$, with $d(L) = \overline{(\log v_{t+L} - \log v_t)^2}$ over 15 log-spaced integer lags in $[1, T/4]$.
- **Leverage**: $\mathrm{lev}(L) = \mathrm{corr}(x_t, v_{t+L})$. The objective uses $L = 1, \dots, 40$; the plots show $L = -40, \dots, 40$.
- **Taylor**: $\mathrm{acf}(|x|, L)$ and $\mathrm{acf}(x^2, L)$ at $L \in \{1, 3, 5, 10, 15, 20\}$, fitted as two curves. The difference is plotted.
- **Zumbach**: the two curves $\mathrm{corr}(p_R^2, f_V)$ and $\mathrm{corr}(p_V, f_R^2)$ at $L \in \{2, 10, 20, 30, 40\}$, fitted separately. Here

$$p_R(t) = \sum_{s=t-L}^{t-1} x_s, \quad f_R(t) = \sum_{s=t}^{t+L-1} x_s, \quad p_V(t) = \frac{1}{L} \sum_{s=t-L}^{t-1} v_s, \quad f_V(t) = \frac{1}{L} \sum_{s=t}^{t+L-1} v_s.$$

  The difference is plotted.
- **Excess kurtosis** of x_t (raw).

Lag-indexed statistics pass through a 3-point moving average across adjacent lags, identically for model and data. Differences are computed per path or per window, so their bands are bands of the difference.

3.3 Targets, tolerances and residual

Real target: the mean over the 6 market windows of the smoothed statistic. Tolerance s_k: the cross-asset standard deviation of the real target over the 9 assets, root-mean-squared over the block's terms.

Table 4: Tolerances s_k (9 assets).

$\hat{H}$	leverage	ACF$\lvert x\rvert$	ACFx^2	$p_R^2 f_V$	$p_V f_R^2$	kurtosis
0.0080	0.0158	0.0309	0.0425	0.0499	0.0637	2.9014

Table 5: Real targets at the first lag of each block (6-window means).

asset	$\hat{H}$	lev(1)	ACF$\lvert x\rvert$(1)	ACFx^2(1)	$p_R^2 f_V(2)$	$p_V f_R^2(2)$	kurtosis
GSPC	0.0701	−0.176	0.212	0.217	0.427	0.314	5.22
RUT	0.0586	−0.167	0.212	0.203	0.354	0.226	3.31
NDX	0.0675	−0.164	0.195	0.217	0.389	0.331	3.50
GPC	0.0495	−0.050	0.183	0.141	0.292	0.231	12.36
QCOM	0.0533	−0.071	0.158	0.105	0.243	0.131	7.04
INTC	0.0561	−0.066	0.159	0.123	0.253	0.159	7.80
LUV	0.0463	−0.084	0.156	0.144	0.308	0.189	5.98
NI	0.0542	−0.084	0.214	0.190	0.321	0.243	8.94
KMB	0.0513	−0.051	0.153	0.121	0.262	0.223	9.03

The residual has 7 blocks and 64 terms, with weights $w_{\text{lev}} = 0.40$, $w_{\hat{H}} = 0.30$, $w_{\text{kurt}} = 0.20$, and 0.025 for each of the four curve blocks. Each block is normalised by $s_k\sqrt{n_k}$, and leverage terms carry the decay $1/(L+1)^2$:

$$
\begin{aligned}
r_{\hat{H}} &= \sqrt{0.30}\,\frac{\bar{\hat{H}} - \hat{H}^{\text{real}}}{s_{\hat{H}}} && \text{(1 term)},\\
r_{\text{lev}}(L) &= \sqrt{0.40}\,\frac{1}{(L+1)^2}\,\frac{\overline{\text{lev}}(L) - \text{lev}^{\text{real}}(L)}{s_{\text{lev}}\sqrt{40}} && L = 1,\dots,40,\\
r_{\text{abs}}(L) &= \sqrt{0.025}\,\frac{\overline{\text{acf}}(|x|,L) - \text{acf}^{\text{real}}(|x|,L)}{s_{\text{abs}}\sqrt{6}} && \text{(6 terms)},\\
r_{\text{sq}}(L) &= \sqrt{0.025}\,\frac{\overline{\text{acf}}(x^2,L) - \text{acf}^{\text{real}}(x^2,L)}{s_{\text{sq}}\sqrt{6}} && \text{(6 terms)},\\
r_{p_R^2 f_V}(L) &= \sqrt{0.025}\,\frac{\overline{\text{corr}}(p_R^2, f_V) - \text{corr}^{\text{real}}(p_R^2, f_V)}{s_{p_R^2 f_V}\sqrt{5}} && \text{(5 terms)},\\
r_{p_V f_R^2}(L) &= \sqrt{0.025}\,\frac{\overline{\text{corr}}(p_V, f_R^2) - \text{corr}^{\text{real}}(p_V, f_R^2)}{s_{p_V f_R^2}\sqrt{5}} && \text{(5 terms)},\\
r_{\text{kurt}} &= \sqrt{0.20}\,\frac{\overline{\text{kurt}} - \text{kurt}^{\text{real}}}{s_{\text{kurt}}} && \text{(1 term)},
\end{aligned}
$$

where the bar is the average over the K calibration paths, and the objective is $Q(\theta) = \|r(\theta)\|_2^2$.

3.4 Simulation, common random numbers and optimiser

Paths are simulated by the exact scheme with 13 sub-steps per day:

- the exact joint law of the Layer-1 increments and ΔW^X, with the OU noise scaled by $\sqrt{\Gamma}$;
- Layer 2 by exponential integration;

- Γ by exact decay plus at most one Exp(a_J) jump per sub-step;
- round(13ϕ) overnight sub-steps, and exact Brownian-bridge extremes in session sub-steps.

Every path starts from the initial state (2.4), with no burn-in. One call evaluates many (asset, θ) blocks of K paths in one vectorised pass. Every block shares the same K noise paths (Gaussian stream seed 11, jump stream seed $11 + 7919$), so common random numbers hold within and across assets.

Algorithm 1 Joint calibration, all assets simultaneously

Require: targets and tolerances; $\theta_a \leftarrow \theta_0$ and damping $\mu_a \leftarrow 10^{-2}$ for every asset a; steps $h = 0.02\,(\mathrm{UB}-\mathrm{LB})$

for iteration $= 1, \ldots, 25$ **do**
  Jacobian pass. One simulation of the blocks $\{(a, \theta_a)\} \cup \{(a, \theta_a \pm h_i e_i)\}_{i=1..17}$ (forward, or backward at the upper bound)
  for each asset a **do** $r_a \leftarrow r(\theta_a)$; $J_a \leftarrow$ finite differences; $Q_a \leftarrow \|r_a\|^2$
  end for
  Step pass. For each a and $\lambda \in \{\mu_a/3, \mu_a, 3\mu_a\}$: $\theta_a^{(\lambda)} \leftarrow \mathrm{clip}_{[\mathrm{LB},\mathrm{UB}]}\big(\theta_a - (J_a^\top J_a + \lambda\,\mathrm{diag} J_a^\top J_a)^{-1} J_a^\top r_a\big)$; one simulation of all 3×9 blocks
  for each asset a **do**
    if $\min_\lambda Q(\theta_a^{(\lambda)}) < Q_a$ **then** accept the best; $\mu_a \leftarrow \lambda_{\mathrm{best}}/2$
    else $\mu_a \leftarrow 4\mu_a$
    end if
  end for
  stop if every asset has improved by less than 10^{-3} (relative) for 3 iterations
end for

Note. The assets are independent problems that share only the noise. The run splits them over two processes, with the same seeds, so the result is identical to a single joint run.

3.5 Validation

For each asset, 20 independent replicate paths of $T = 1\,342$ days (seed 99) are simulated at the calibrated θ_a. For each statistic the validation reports:

- the replicate mean;
- the 90% band (5%–95% quantiles across replicates);
- the coverage, i.e. the fraction of lags where the real target lies in the band (Y/N for scalars).

Q is also recomputed on these paths (Q_{val}). As a reference for out-of-sample improvement, Q_{val} is also computed at the common start θ_0.

3.6 Number of calibration paths

A first run used $K = 16$ paths, as in the previous version of this calibration, and overfitted. With 17 parameters, the initial state among them, the optimiser fitted the common noise of the 16 paths. Q_{cal} fell to 0.04–0.27, but Q_{val} was 0.73–1.11. For five stocks, Q_{val} was even worse than at the common start: for QCOM, 0.81 against 0.15 at θ_0. The excess came almost entirely from the $\hat{H}$ block, which was matched on the calibration paths only. Its tolerance, 0.008, is of the order of the Monte Carlo error of a 16-path mean. The run reported here uses $K = 64$.

4 Results

4.1 Calibrated parameters

Table 6: Calibrated parameters († = at a bound of Table 3), overnight share ϕ, and mean annualised GK volatility (real / model, validation paths).

	GSPC	RUT	NDX	GPC	QCOM	INTC	LUV	NI	KMB
c_1	$0.300^\dagger$	$0.300^\dagger$	$0.300^\dagger$	0.648	0.555	0.515	0.581	0.491	0.668
c_2	0.567	0.345	0.542	0.568	0.430	0.491	0.457	0.517	0.513
c_3	$0.650^\dagger$	$0.650^\dagger$	$0.650^\dagger$	0.381	0.328	0.425	0.278	0.428	0.285
c_4	$0.600^\dagger$	$0.600^\dagger$	$0.600^\dagger$	$0.200^\dagger$	0.439	0.413	$0.200^\dagger$	0.215	0.208
c_5	$0.600^\dagger$	$0.600^\dagger$	$0.600^\dagger$	$0.200^\dagger$	$0.600^\dagger$	0.526	$0.200^\dagger$	$0.200^\dagger$	0.235
d_1	$0.000^\dagger$	$0.300^\dagger$	$0.000^\dagger$	$0.300^\dagger$	$0.300^\dagger$	0.277	$0.300^\dagger$	$0.300^\dagger$	0.294
d_2	$0.000^\dagger$	$0.300^\dagger$	$0.000^\dagger$	$0.300^\dagger$	$0.300^\dagger$	0.158	$0.300^\dagger$	0.278	0.291
d_3	0.094	0.069	0.041	0.215	0.099	0.116	0.040	0.135	0.075
d_4	0.140	$0.000^\dagger$	0.047	$0.000^\dagger$	$0.000^\dagger$	0.054	$0.000^\dagger$	$0.000^\dagger$	$0.000^\dagger$
e_1	$0.000^\dagger$	$0.000^\dagger$	0.019	$0.000^\dagger$	$0.000^\dagger$	$0.000^\dagger$	$0.000^\dagger$	$0.000^\dagger$	$0.000^\dagger$
a	$0.150^\dagger$	$0.250^\dagger$	$0.250^\dagger$	0.177	0.193	$0.250^\dagger$	$0.250^\dagger$	$0.150^\dagger$	$0.250^\dagger$
b	$0.500^\dagger$	$5.000^\dagger$	1.269	$0.500^\dagger$	$5.000^\dagger$	2.748	2.743	$0.500^\dagger$	2.257
c	$0.250^\dagger$	0.152	$0.150^\dagger$	0.194	$0.250^\dagger$	$0.250^\dagger$	$0.250^\dagger$	$0.250^\dagger$	$0.150^\dagger$
ϱ	0.673	$0.700^\dagger$	0.675	0.465	$0.700^\dagger$	$0.700^\dagger$	$0.700^\dagger$	$0.700^\dagger$	0.611
ζ_0	-1.264	$-1.500^\dagger$	$-1.500^\dagger$	$1.500^\dagger$	0.983	0.483	-1.004	$1.500^\dagger$	0.272
P_0	$1.500^\dagger$	$1.500^\dagger$	$-1.500^\dagger$	$-1.500^\dagger$	$-1.500^\dagger$	$1.500^\dagger$	$1.500^\dagger$	$-1.500^\dagger$	$1.500^\dagger$
Γ_0	$0.300^\dagger$	$1.500^\dagger$	0.347	$1.500^\dagger$	$1.500^\dagger$	0.530	$1.500^\dagger$	$1.500^\dagger$	$0.300^\dagger$
ϕ	0.357	0.464	0.347	0.105	0.227	0.313	0.137	0.057	0.133
vol real	0.114	0.124	0.171	0.185	0.320	0.270	0.286	0.178	0.168
vol model	0.255	0.160	0.193	0.217	0.240	0.245	0.303	0.239	0.171
at a bound	12/17	14/17	10/17	10/17	10/17	5/17	11/17	11/17	6/17

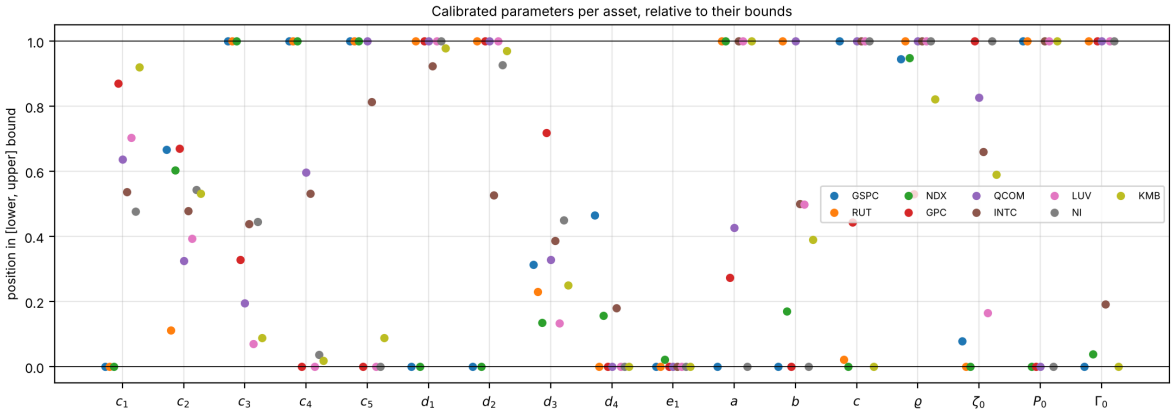


Figure 1: Calibrated parameters relative to their bounds.

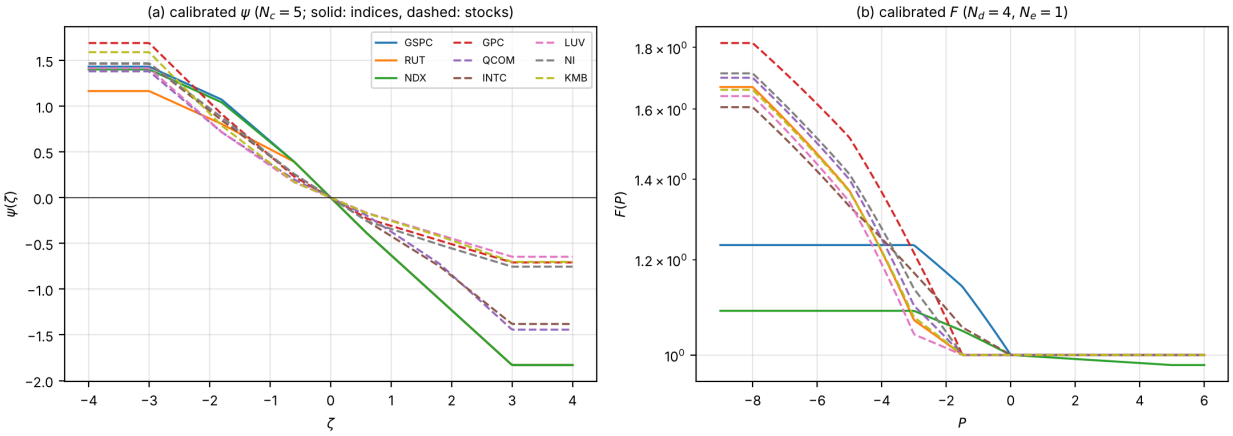


Figure 2: Calibrated readouts per asset: ψ (5 pieces) and F (4 loss-side pieces, 1 gain-side piece).

- **Readout ψ.**
  - The indices take the largest slopes on the positive side ($c_4 = c_5 = 0.6$, at the bound): deep calm spells.
  - The stocks take steep negative-side slopes (c_1 up to 0.67) and flatter positive-side slopes ($c_4, c_5 \approx 0.2$, several at the bound).
  - This is the separation of the previous version: indices ask for more downside in volatility, stocks for spikes.
- **Feedback F.**
  - The loss side is active for every asset; the gain side is switched off ($e_1 \approx 0$ for all nine assets, at most 0.02).
  - For the Russell 2000 and five of the six stocks (all but INTC) the increments d_1, d_2 sit at or near their upper bound, i.e. F rises fastest at the far loss side ($P < -5$), up to $F(-8) \approx 1.6$–1.8.
  - For the S&P 500 and Nasdaq-100, F is active only near 0 (d_3, d_4) and stays below 1.25.
- **Leverage correlation.** ϱ is at or near its upper bound 0.7 for seven assets.
- **Level, term structure and initial state.** These end on bounds for most assets, as Proposition 3.1 predicts. The model's mean volatility differs from the market's by up to a factor of 2.2 (S&P 500: 0.255 vs. 0.114).

4.2 Objective and convergence

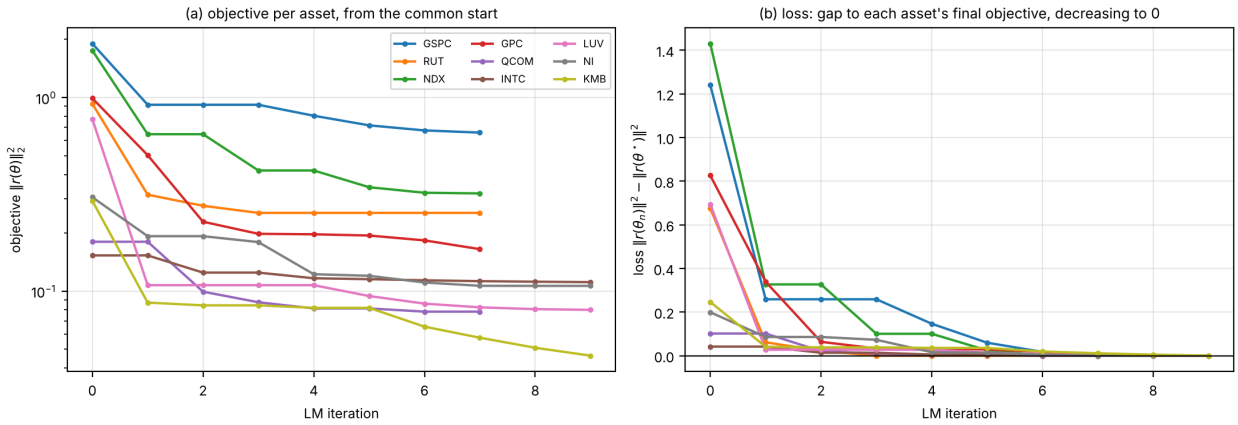


Figure 3: (a) Objective per asset from the common start ($K = 64$). (b) Loss, the gap to each asset's last objective, decreasing to 0.

Table 7: Objective: at the common start and after calibration ($K = 64$, seed 11); on the 20 validation replicates at the common start θ_0 and at the calibrated θ; the validation objective of the previous version (two-slope readout, unmodulated bank); and the block contributions on validation.

	Q_{start}	Q_{cal}	$Q_{\text{val}}(\theta_0)$	Q_{val}	previous	$\hat{H}$	lev	ACF$\lvert x\rvert$	ACFx^2	$p_R^2 f_V$	$p_V f_R^2$	kurt
GSPC	1.899	0.659	2.747	**1.319**	0.467	0.625	0.003	0.179	0.146	0.281	0.075	0.009
RUT	0.929	0.253	1.093	**0.368**	0.438	0.072	0.004	0.152	0.089	0.044	0.003	0.004
NDX	1.750	0.319	2.407	**0.594**	0.445	0.267	0.007	0.076	0.105	0.083	0.040	0.016
GPC	0.991	0.165	1.132	**0.249**	0.960	0.002	0.021	0.061	0.018	0.071	0.022	0.055
QCOM	0.180	0.078	0.145	**0.309**	0.448	0.077	0.015	0.044	0.003	0.069	0.001	0.101
INTC	0.153	0.111	0.341	**0.346**	0.430	0.082	0.021	0.052	0.007	0.078	0.006	0.100
LUV	0.774	0.080	0.440	**0.245**	0.559	0.044	0.013	0.053	0.012	0.083	0.011	0.029
NI	0.305	0.106	0.521	**0.412**	0.864	0.057	0.016	0.097	0.043	0.088	0.021	0.091
KMB	0.293	0.046	0.308	**0.327**	0.357	0.139	0.019	0.016	0.007	0.026	0.012	0.109

1. **In sample.** The objective falls by a factor of 1.4–9.7 from the common start.

2. **Out of sample.**

   - Q_{val} improves on the common start for the three indices (GSPC 2.75 → 1.32, RUT 1.09 → 0.37, NDX 2.41 → 0.59), GPC (1.13 → 0.25), LUV (0.44 → 0.24) and NI (0.52 → 0.41).
   - It is about unchanged for INTC (0.34) and KMB (0.31 → 0.33), and worse for QCOM (0.15 → 0.31). For these three assets the common start was already close to the targets, and the residual overfitting is of the order of the gain.

3. **Against the previous version.** Q_{val} is lower for seven of the nine assets, most for GPC (0.25 vs. 0.96) and NI (0.41 vs. 0.86). It is higher for the S&P 500 (1.32 vs. 0.47) and the Nasdaq-100 (0.59 vs. 0.45), whose largest blocks are $\hat{H}$ and the Taylor and Zumbach curves.

4. **Convergence.** The run was stopped after 7 and 9 iterations, with the objective still decreasing by about 1–3% per iteration for some assets. A completed run of 25 iterations would lower Q_{cal} somewhat further.

4.3 Roughness

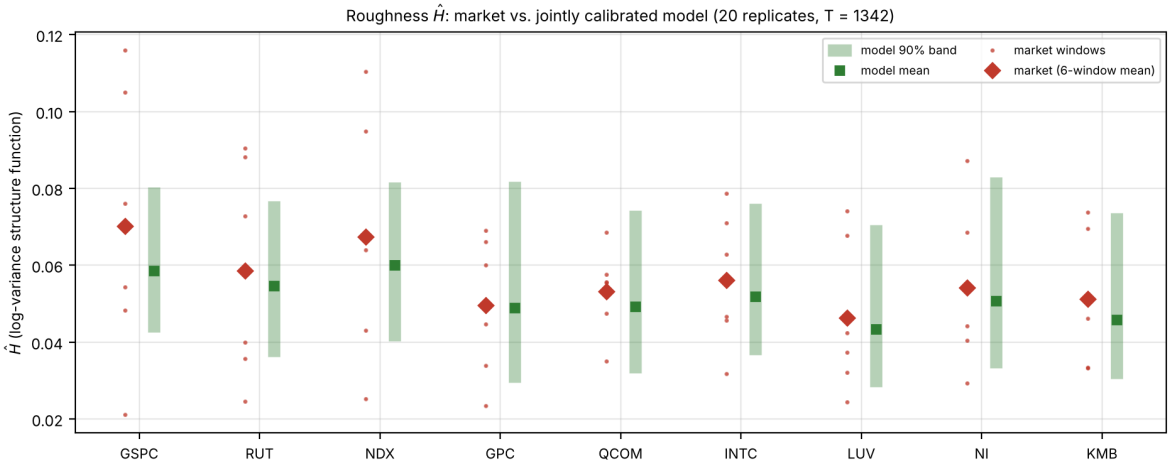


Figure 4: $\hat{H}$: real 6-window mean (red diamond) and windows (red dots); model mean (green square) and 90% band, 20 replicates.

The real $\hat{H}$ lies inside the band for all nine assets. Unlike the previous version, whose model was 0.005–0.013 too rough, the model mean is now 0.0006–0.011 *below* the market. The deficit is largest for the S&P 500 (0.059 vs. 0.070) and the Nasdaq-100 (0.060 vs. 0.068). The architecture's $H = 0.058$ is shared, and the readout alone cannot raise $\hat{H}$ to the indices' values.

4.4 Leverage, lags -40 to 40

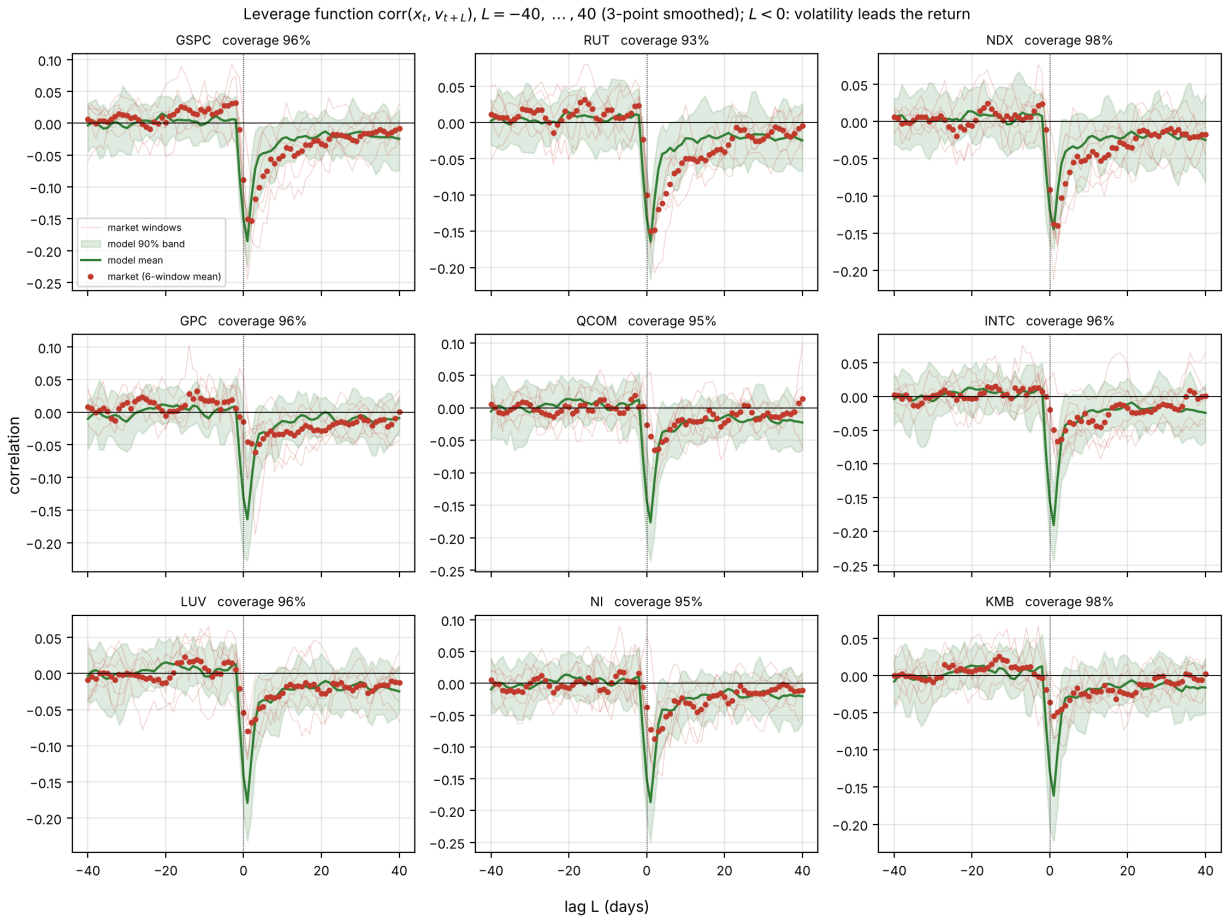


Figure 5: Leverage $\mathrm{corr}(x_t, v_{t+L})$ for $L = -40, \ldots, 40$ (3-point smoothed).

- **Negative lags.** The model reproduces the causal asymmetry: no correlation when volatility leads the return.
- **Positive lags.** The model follows the market; coverage is 85–97%.
- **Lag 0**, not part of the objective. The same-day correlation is too strong. For the indices the model gives -0.12 to -0.15, against -0.09 to -0.10 in the market. For the stocks it gives -0.13 to -0.16, against -0.02 to -0.05. This comes from ϱ near 0.7 together with the 37-day leverage window: the window concentrates the leverage loading on the slower rough modes, so a large ϱ is needed to reach the lag-1 target.

Coverage over lags -40 to 40 is 93–98% (average 96%).

4.5 Taylor effect

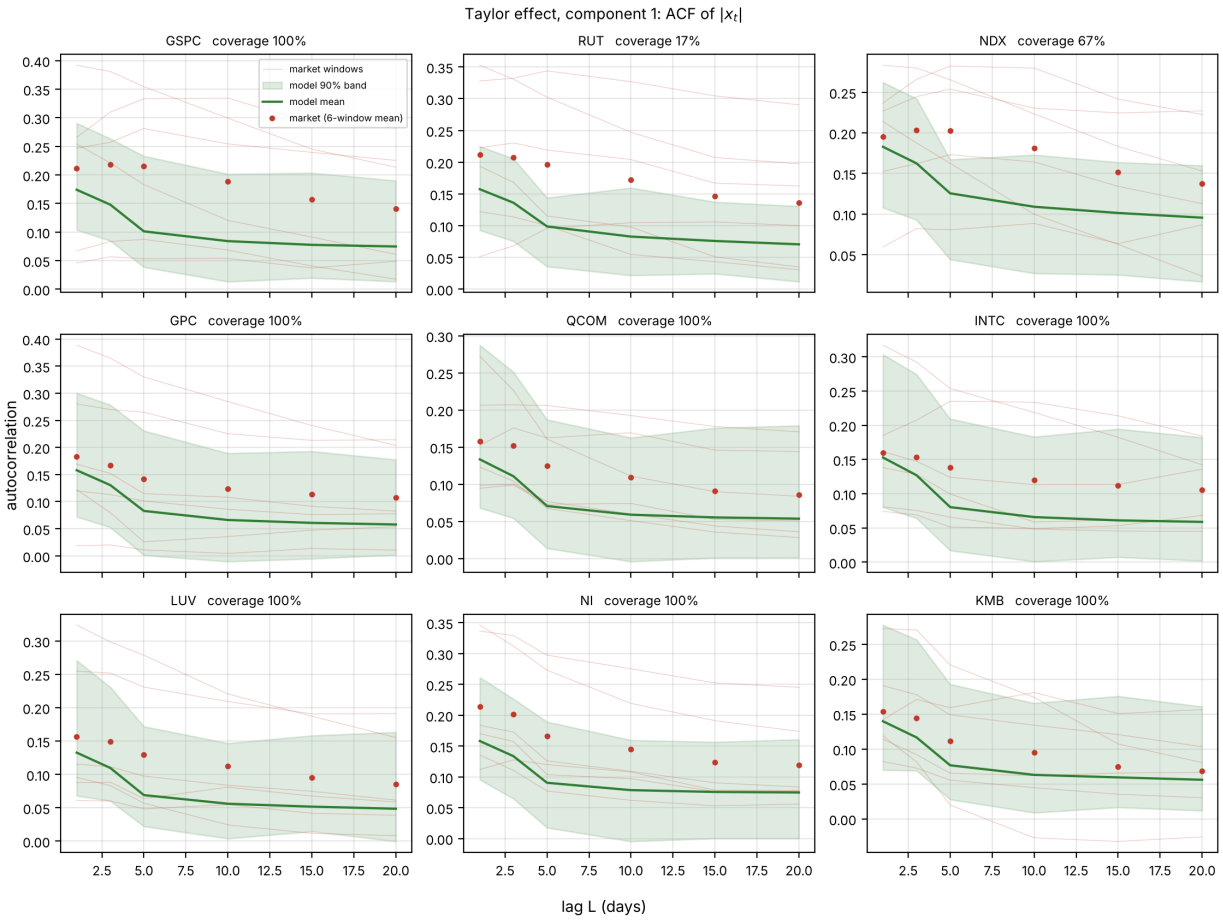


Figure 6: Taylor component 1: ACF of $|x_t|$.

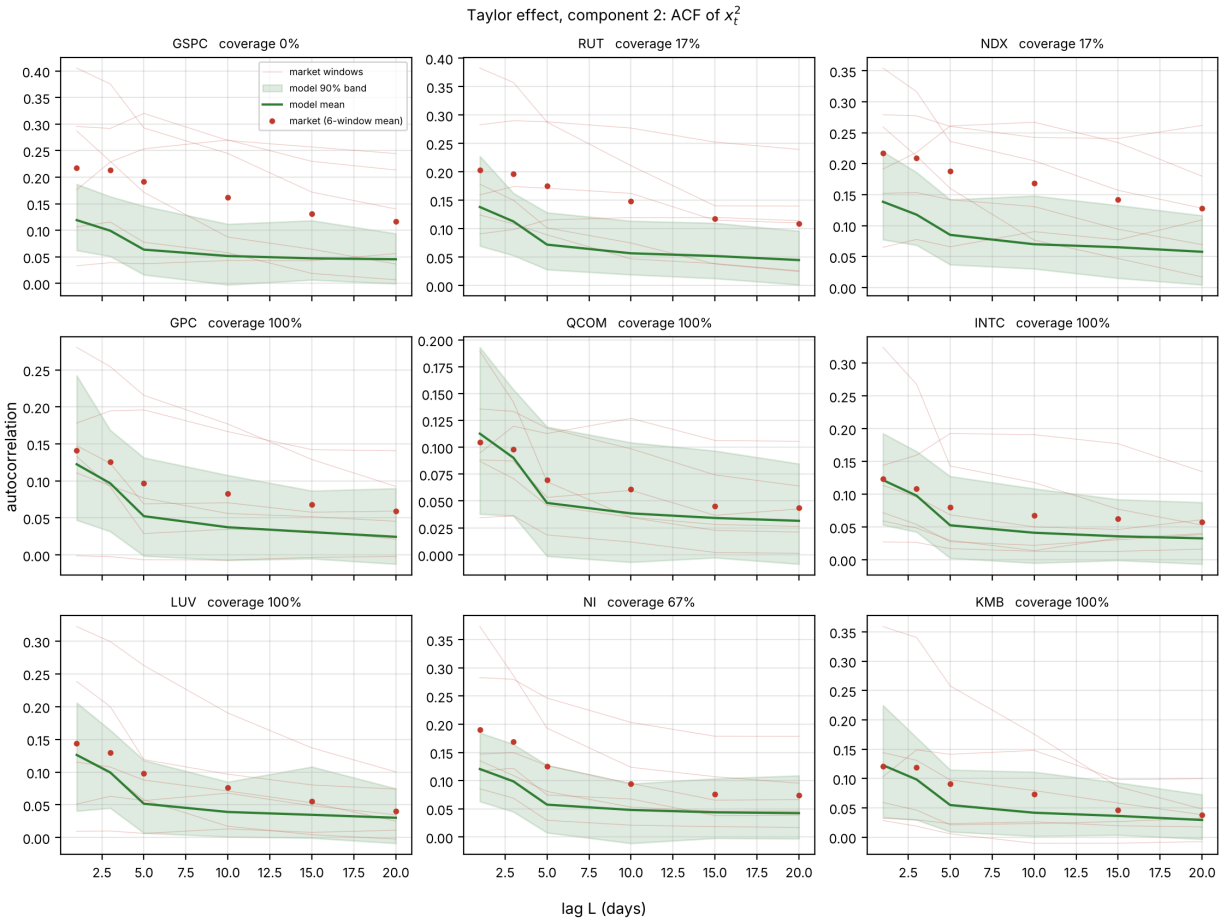


Figure 7: Taylor component 2: ACF of x_t^2.

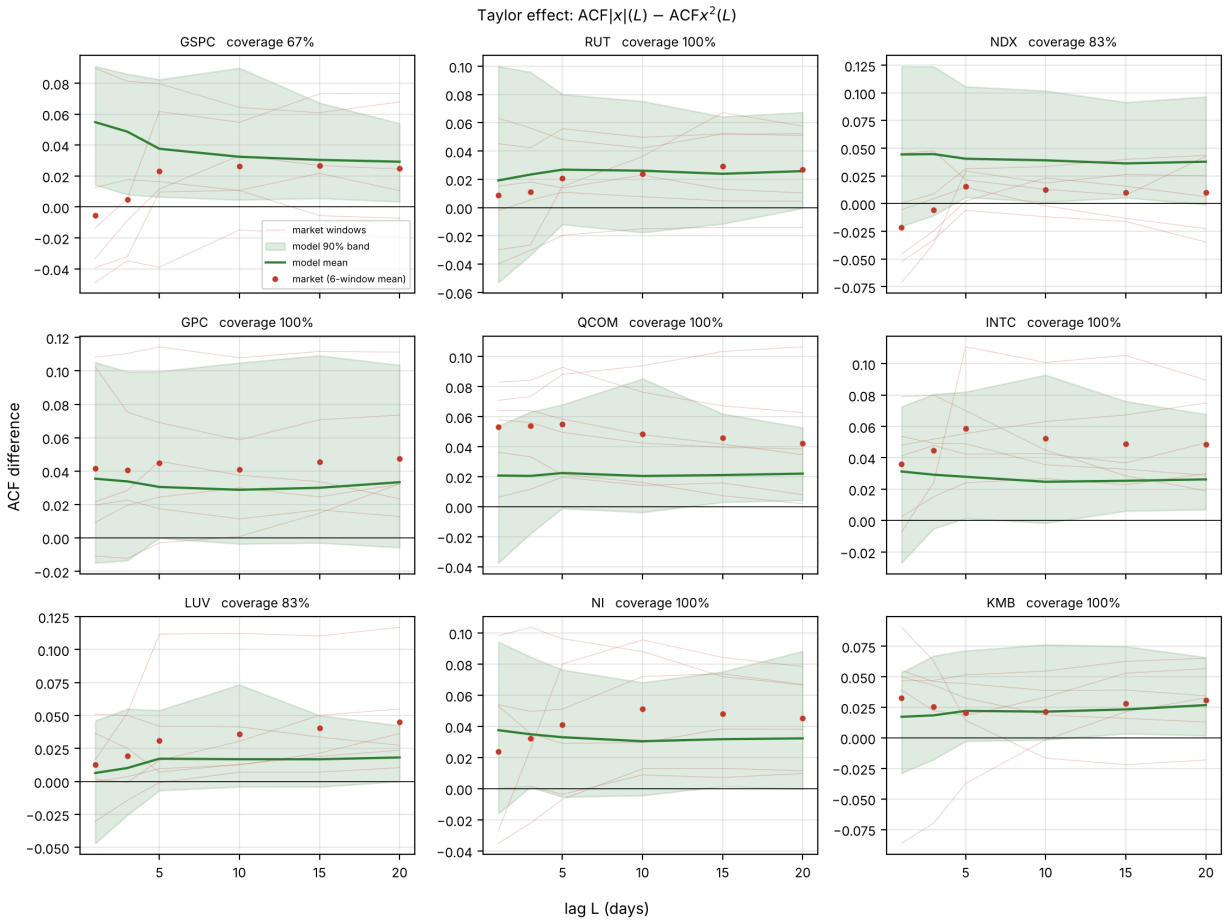


Figure 8: Taylor effect: ACF$|x|$ − ACFx^2.

- **Stocks and Russell 2000.** The difference is reproduced in sign and order: model 0.014–0.033 vs. market 0.020–0.050, with coverage 83–100%. For the stocks it is now slightly too small; in the previous version it was matched.
- **S&P 500 and Nasdaq-100.** As in the previous version, the model's ACFx^2 is too low and the difference too large: 0.039 and 0.041 vs. 0.017 and 0.004. Large index moves cluster more than the model's.

4.6 Zumbach effect

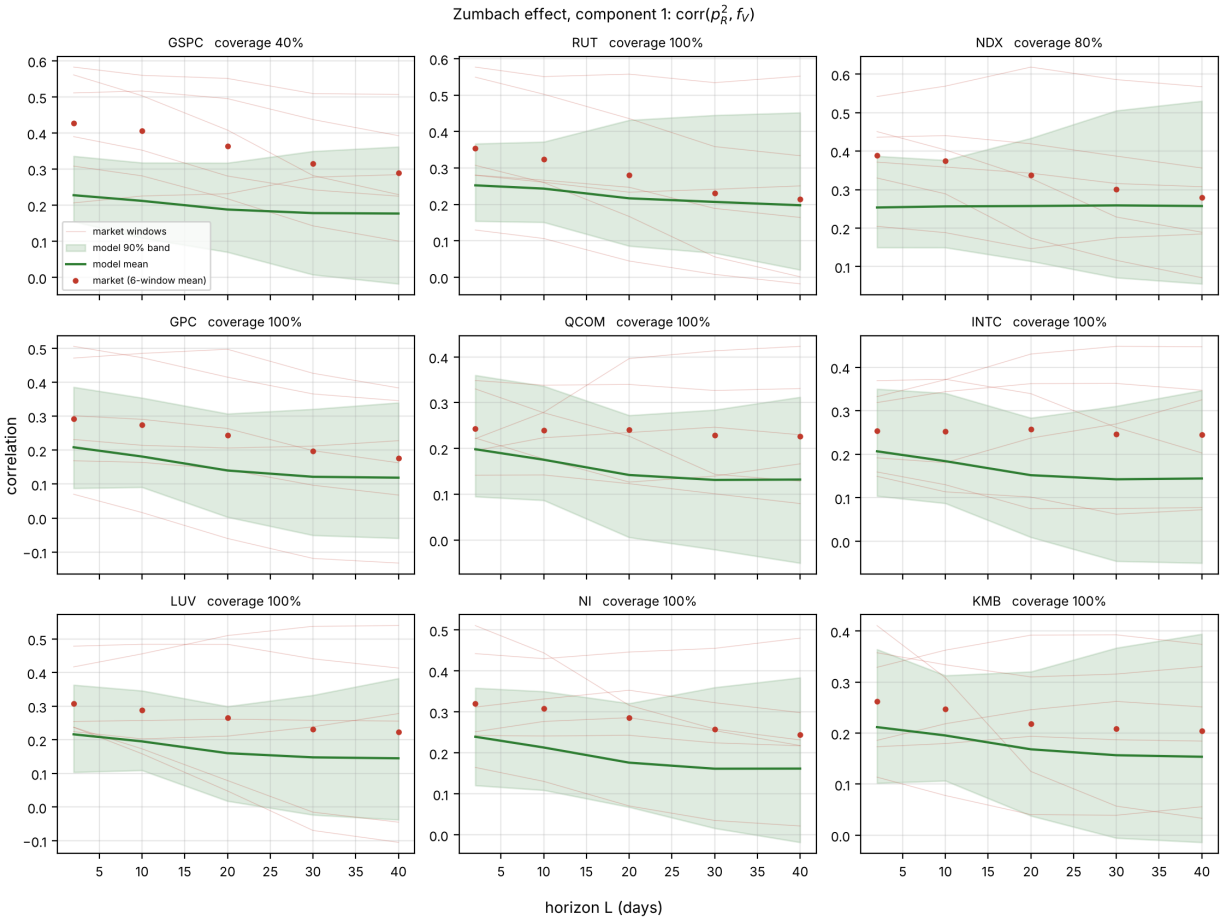


Figure 9: Zumbach component 1: $\mathrm{corr}(p_R^2, f_V)$.

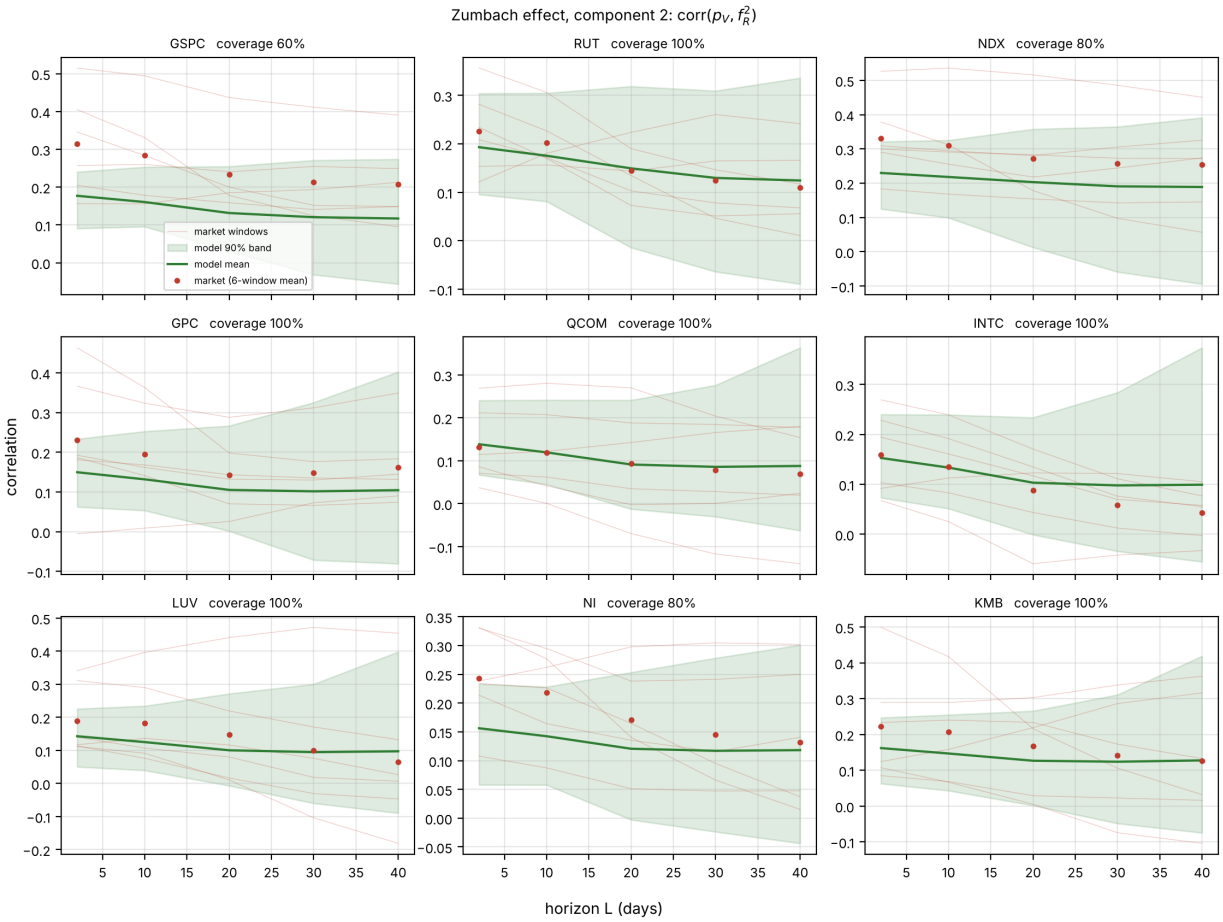


Figure 10: Zumbach component 2: $\mathrm{corr}(p_V, f_R^2)$.

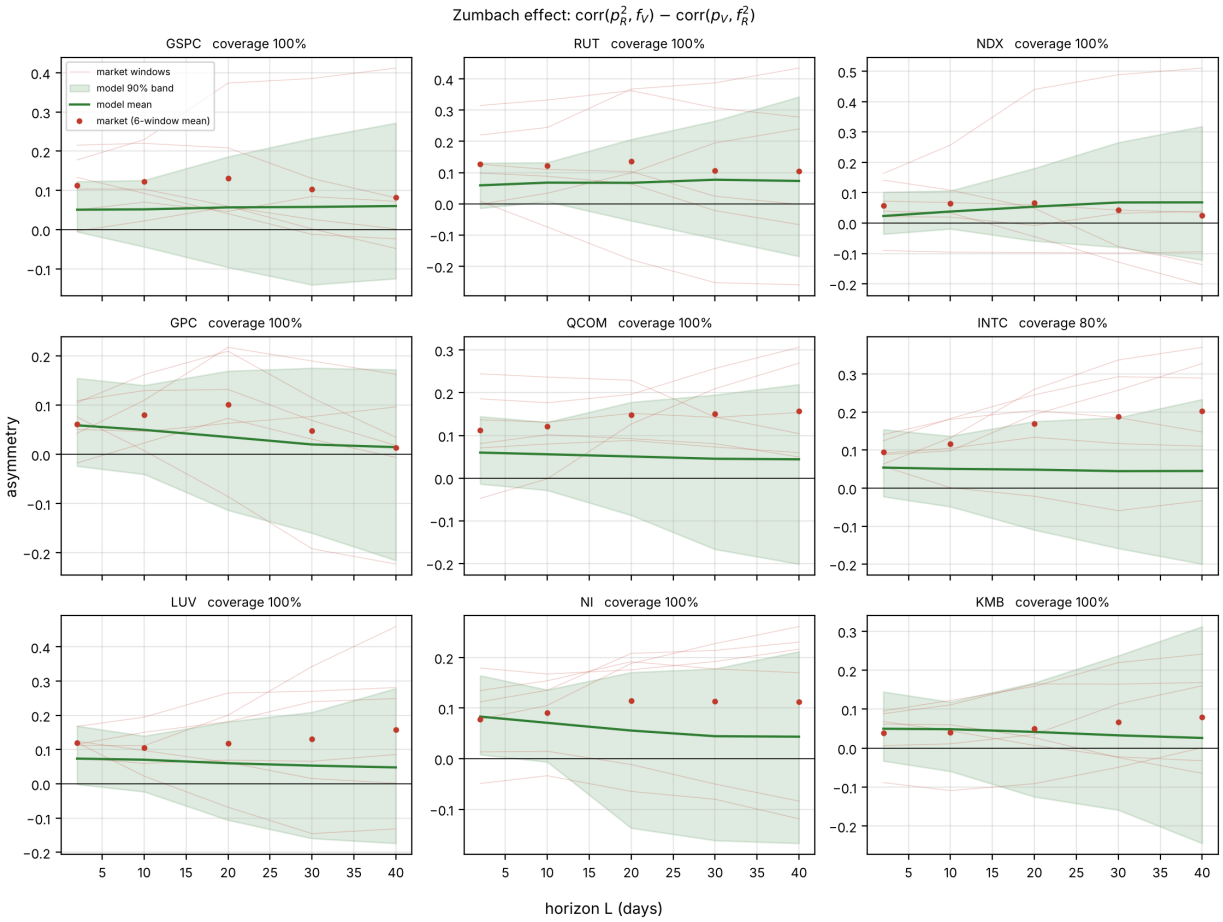


Figure 11: Zumbach effect: $\mathrm{corr}(p_R^2, f_V) - \mathrm{corr}(p_V, f_R^2)$.

- **Stocks.** The mean asymmetry over horizons is 0.036–0.061, against 0.055–0.154 in the market. This is still too weak, but clearly larger than in the previous version (−0.011–0.021). The loss-side feedback F and the modulation both contribute.
- **Indices.** NDX is matched (0.050 vs. 0.052). GSPC and RUT are too weak (0.056 and 0.069 vs. 0.110 and 0.119).

Coverage of the difference is 80–100% (average 98%).

4.7 Return distribution and sample paths

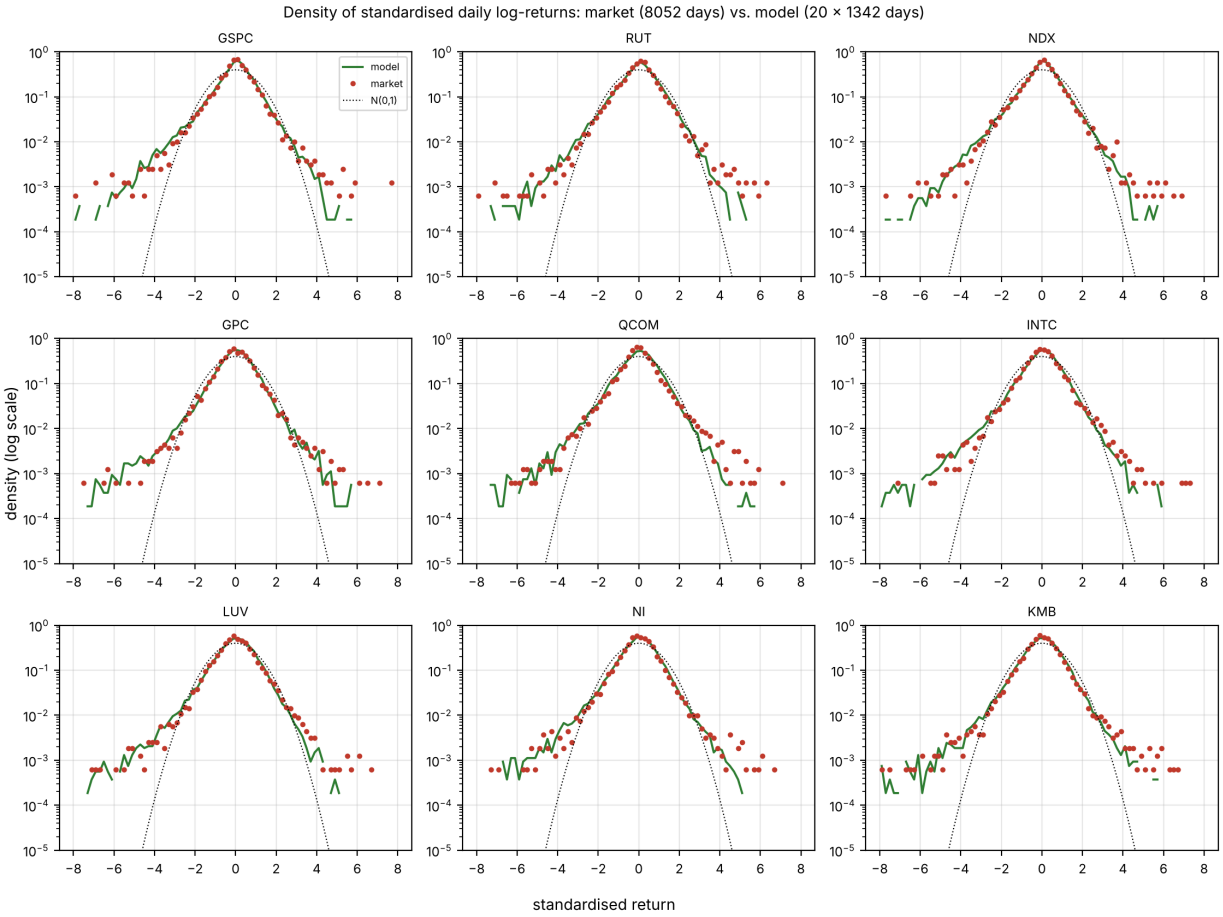


Figure 12: Density of standardised daily log-returns (log scale).

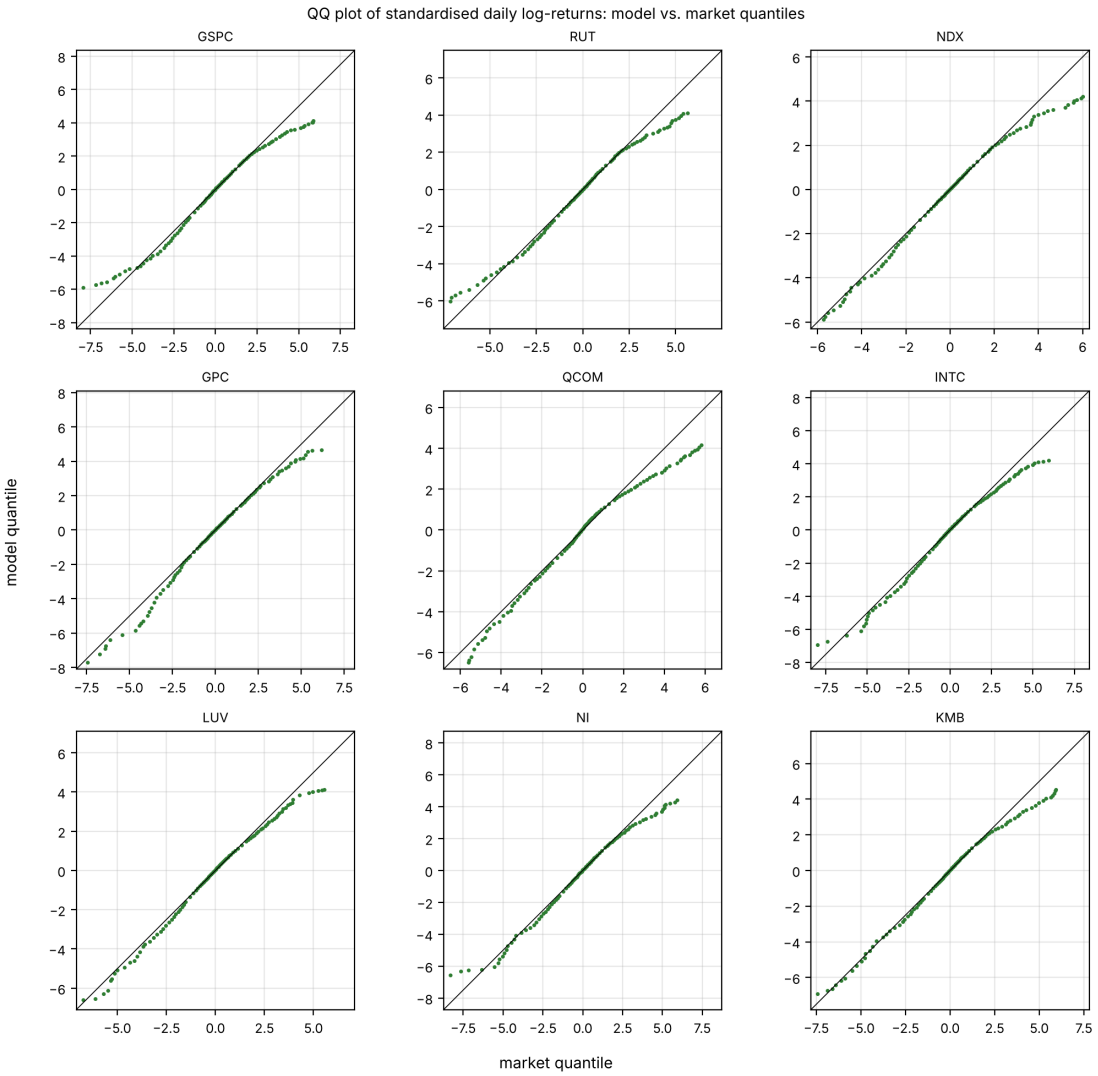


Figure 13: QQ plot of standardised daily log-returns.

Kurtosis is inside the band for all nine assets. The model mean is 3.7–10.8, against market 3.3–12.4, below the market for the six stocks (e.g. KMB 6.9 vs. 9.0).

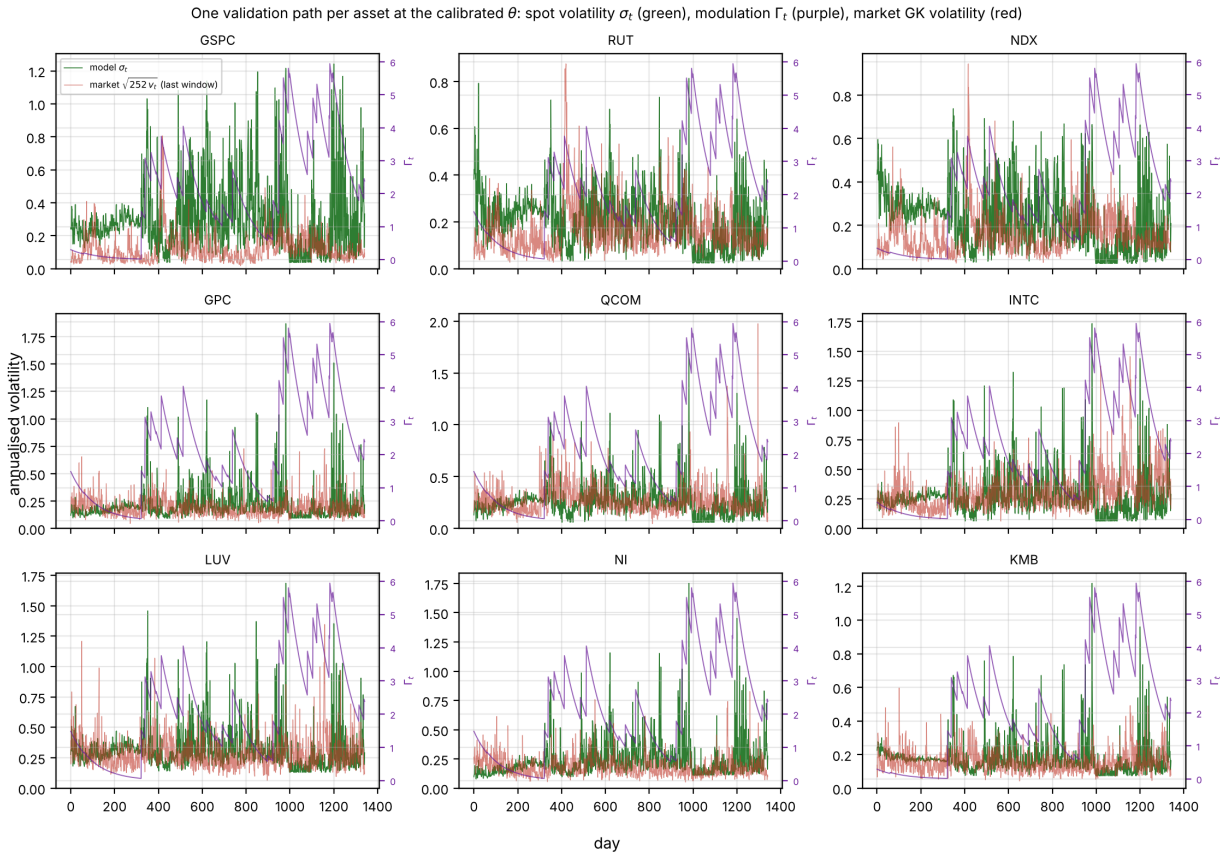


Figure 14: One validation path per asset: σ_t, Γ_t and the market's GK volatility over its last window. The paths share the same noise, including the jumps of Γ (common random numbers).

4.8 Coverage

Table 8: Coverage of the 7 calibrated blocks: fraction of lags (Y/N for scalars) where the real target lies in the model's 90% band, 20 replicates.

asset	$\hat{H}$	leverage	ACF$\lvert x\rvert$	ACFx^2	corr(p_R^2, f_V)	corr(p_V, f_R^2)	kurtosis
GSPC	Y	98%	100%	0%	40%	60%	Y
RUT	Y	85%	17%	17%	100%	100%	Y
NDX	Y	92%	67%	17%	80%	80%	Y
GPC	Y	95%	100%	100%	100%	100%	Y
QCOM	Y	95%	100%	100%	100%	100%	Y
INTC	Y	98%	100%	100%	100%	100%	Y
LUV	Y	95%	100%	100%	100%	100%	Y
NI	Y	95%	100%	67%	100%	80%	Y
KMB	Y	95%	100%	100%	100%	100%	Y
average	100%	94%	87%	67%	91%	91%	100%

Table 9: Coverage of the additional plotted quantities.

asset	leverage, $L = -40..40$	Taylor difference	Zumbach difference
GSPC	96%	67%	100%
RUT	93%	100%	100%
NDX	98%	83%	100%
GPC	96%	100%	100%
QCOM	95%	100%	100%
INTC	96%	100%	80%
LUV	96%	83%	100%
NI	95%	100%	100%
KMB	98%	100%	100%
average	96%	93%	98%

For comparison, the previous version reports average coverage of:

- $\hat{H}$ and kurtosis 100%;
- leverage 98%;
- ACF$|x|$ 100%, ACFx^2 78%;
- Zumbach 98% and 100%.

Coverage is a weak criterion with 20 replicates of 1 342 days; the mean curves and Q_{val} are the sharper measures.

5 Limitations and next steps

1. **Incomplete run.** The calibration was stopped after 7–9 of its 25 iterations. Running `reproduce.sh` to completion would lower Q_{cal} further.
2. **Residual overfitting.** Even with $K = 64$, three stocks (QCOM, INTC, KMB) do not improve out of sample on the common start. $K = 128$, or fixing the weakly identified parameters (level and initial state), would reduce it.
3. **S&P 500 and Nasdaq-100.** Roughness is too low, the Taylor difference too large and the Zumbach asymmetry too weak. A per-asset H would address the first, since the architecture is shared and H is a hyperparameter.
4. **Same-day leverage too strong** for all assets. A lag-0 term in the objective would constrain ϱ.
5. **The box is binding.** Many readout parameters sit at the bounds of the grid-search levels; the indices ask for more positive-side slope and the stocks for more loss-side feedback. Widening the box is the natural next test.
6. **Level not identified** (Proposition 3.1). A mean-volatility target would match the model's level to the market's.

A Code reference

All files carry the banner docstring. Run `reproduce.sh` inside `code/`; `DATA_DIR` must contain the two data files.

file	content
`jc_core.py`	data, GK proxy, overnight shares, statistics, smoothing, targets and tolerances; parameter box, readouts (`psi_values`, `F_values`, `interp_rows`); architecture (`load_architecture`); batched exact simulator with common random numbers, modulated bank, initial state and measurement layer (`simulate_batch`); residual
`jc_calibrate.py`	Algorithm 1; `python jc_calibrate.py 25 ASSETS tag`; `merge` combines the parts; `K_CAL` (default 64)
`jc_finalize.py`	builds `calibration.json` from the per-iteration progress files when the run is stopped early (used for the results reported here)
`jc_evaluate.py`	20-replicate validation, bands and coverage, two-sided leverage, Taylor and Zumbach differences, standardised returns
`jc_figs.py`	Figures 1–10
`jc_export.py`	`targets.json`, `summary.json`
`architecture.json`	the fixed architecture (grid-search optimum)
`reproduce.sh`	the full pipeline (about 45 minutes on 2 cores)

Note. `simulate_batch` takes one row of 17 parameters per block. Every block reuses the same K Gaussian and jump draws, so finite-difference probes differ only through θ. Because the readouts are piecewise linear with per-path knot values, they are evaluated row-wise by `interp_rows`.

Results (JSON):

- `calibration.json`: θ per asset, Q, block contributions, iteration history;
- `calibration_progress_p1.json`, `calibration_progress_p2.json`: per-iteration state of the two processes;
- `evaluation.json`: per asset and statistic, the model mean, band, target, windows and coverage;
- `summary.json`: architecture, parameters with knot values of ψ and F, parameters at a bound, objectives, coverage, mean volatility;
- `targets.json`; `returns_std.npz` (standardised returns).